

Regular Expressions



Matching Characters

.	Matches	any character except newline
\w	Matches	any word character (alphanumeric + underscore)
\W	Matches	any non-word character
\d	Matches	any digit (0-9)
\D	Matches	any non-digit
\s	Matches	any whitespace character (space, tab, newline)
\S	Matches	any non-whitespace character



Groups and Ranges

(abc)	Capturing group	- captures the matched text
(?:abc)	Non-capturing group	- doesn't capture the match
	Matches	either the expression before or after the pipe
\1 \2 \3	References	to captured groups (1st, 2nd, 3rd)

Anchors

^	Matches	the start of a string
\$	Matches	the end of a string
\b	Matches	a word boundary
\B	Matches	a non-word boundary



Escaping special characters

\	Escapes	a special character (eg \ matches a literal period)
[\\^\$. ?*(+){}		Need escaping to match literally

Quantifiers

*	Matches	0 or more of the preceding
+	Matches	1 or more of the preceding element
?	Matches	0 or 1 of the preceding element
{n}	Matches	exactly n occurrences
{n,}	Matches	n or more occurrences



Flag / Modifiers

g	Global	find all matches rather than stopping after the first
i	Case-insensitive	ignore case when matching
m	Multiline	^ and \$ match start/end of each line
s	Dotall	allows . to match newlines too
x	Extended	allows comments and whitespace in pattern

Greedy vs Lazy Quantifiers

* Greedy	matches as many characters as possible
*? Lazy:	matches as few characters as possible
+? ?? {n}?	Lazy versions of +, ? and {n}



Unicode Properties (with u flag)

\p{L}	Matches	any letter from any language
\p{N}	Matches	any kind of numeric character
\p{Sc}	Matches	any currency symbol



Character Classes

[abc]	Matches	any one of a, b, or c
[^abc]	Matches	any character except a, b, or c
[a-z]	Matches	any character from a to z
[0-9]	Matches	any digit from 0 to 9



Lookaround Assertions

(?=abc)	Positive lookahead	matches if followed by abc
(?!abc)	Negative lookahead	matches if not followed by abc
(?<=abc)	Positive lookbehind	matches if preceded by abc